

IT6500: Cloud Computing

for postgraduate programme

Lecture 5

Intro. AWS

Dr. Nguyen Binh Minh

Virtual-* Allows for the Scale of Abstraction to Increase Over Time

Run one process within certain resource limits.

Op Sys has virtual memory, virtual CPU, and virtual storage (file system).

Run multiple processes within certain resource limits.

Resource containers (Solaris), virtual servers (Linux), virtual images (Docker)

Run an entire operating system within certain limits.

Virtual machine technology: VMWare, Xen, KVM, etc.

Run a set of virtual machines connected via a private network.

Virtual networks (SDNs) provision bandwidth between virtual machines.

Run a private virtual architecture for every customer.

Automated tools replicate virtual infrastructure as needed.

Amazon AWS

Grew out of Amazon's need to rapidly provision and configure machines of standard configurations for its own business.

Early 2000s – Both private and shared data centers began using virtualization to perform “server consolidation”

2003 – Internal memo by Chris Pinkham describing an “infrastructure service for the world.”

2006 – S3 first deployed in the spring, EC2 in the fall

2008 – Elastic Block Store available.

2009 – Relational Database Service

2012 – DynamoDB

Does it turn a profit?



Amazon Web Services

Compute & Networking

- Direct Connect**
Dedicated Network Connection to AWS
- EC2**
Virtual Servers in the Cloud
- Route 53**
Scalable Domain Name System
- VPC**
Isolated Cloud Resources

Storage & Content Delivery

- CloudFront**
Global Content Delivery Network
- Glacier**
Archive Storage in the Cloud
- S3**
Scalable Storage in the Cloud
- Storage Gateway**
Integrates On-Premises IT Environments with Cloud Storage

Database

- DynamoDB**
Predictable and Scalable NoSQL Data Store
- ElastiCache**
In-Memory Cache
- RDS**
Managed Relational Database Service
- Redshift**
Managed Petabyte-Scale Data Warehouse Service

Deployment & Management

- CloudFormation**
Templated AWS Resource Creation
- CloudTrail**
User Activity and Change Tracking
- CloudWatch**
Resource and Application Monitoring
- Directory Service**
Managed Directories in the Cloud
- Elastic Beanstalk**
AWS Application Container
- IAM**
Secure AWS Access Control
- OpsWorks**
DevOps Application Management Service
- Trusted Advisor**
AWS Cloud Optimization Expert

Analytics

- Data Pipeline**
Orchestration for Data-Driven Workflows
- Elastic MapReduce**
Managed Hadoop Framework
- Kinesis**
Real-time Processing of Streaming Big Data

Mobile Services

- Cognito**
User Identity and App Data Synchronization
- Mobile Analytics**
Understand App Usage Data at Scale
- SNS**
Push Notification Service

App Services

- AppStream**
Low Latency Application Streaming
- CloudSearch**
Managed Search Service
- Elastic Transcoder**
Easy-to-use Scalable Media Transcoding
- SES**
Email Sending Service
- SQS**
Message Queue Service
- SWF**
Workflow Service for Coordinating Application Components

Applications

- WorkSpaces**
Desktops in the Cloud
- Zocalo**
Secure Enterprise Storage and Sharing Service

Additional Resources

Getting Started

See our documentation to get started and learn more about how to use our services.

AWS Console Mobile App

View your resources on the go with our AWS Console mobile app, available from [Amazon Appstore](#), [Google Play](#), or [iTunes](#).

AWS Marketplace

Find and buy software, launch with 1-Click and pay by the hour.

Service Health

✓ All services operating normally.

Updated: Oct 26 2014 21:31:00 GMT-0400

[Service Health Dashboard](#)

Set Start Page

Console Home ▾

Terminology

Instance = One running virtual machine.

Instance Type = hardware configuration: cores, memory, disk.

Instance Store Volume = Temporary disk associated with instance.

Image (AMI) = Stored bits which can be turned into instances.

Key Pair = Credentials used to access VM from command line.

Region = Geographic location, price, laws, network locality.

Availability Zone = Subdivision of region the is fault-independent.

Model	vCPU	CPU Credits / hour	Mem (GiB)	Storage (GB)
t2.micro	1	6	1	EBS Only
t2.small	1	12	2	EBS Only
t2.medium	2	24	4	EBS Only

Model	vCPU	Mem (GiB)	SSD Storage (GB)
c3.large	2	3.75	2 x 16
c3.xlarge	4	7.5	2 x 40
c3.2xlarge	8	15	2 x 80
c3.4xlarge	16	30	2 x 160
c3.8xlarge	32	60	2 x 320

Use Cases

High performance front-end fleets, web-servers, on-demand batch processing, distributed analytics, high performance science and engineering applications, ad serving, batch processing, MMO gaming, video encoding, and distributed analytics.

Model	vCPU	Mem (GiB)	SSD Storage (GB)
m3.medium	1	3.75	1 x 4
m3.large	2	7.5	1 x 32
m3.xlarge	4	15	2 x 40
m3.2xlarge	8	30	2 x 80

Model	vCPU	Mem (GiB)	SSD Storage (GB)
r3.large	2	15.25	1 x 32
r3.xlarge	4	30.5	1 x 80
r3.2xlarge	8	61	1 x 160
r3.4xlarge	16	122	1 x 320
r3.8xlarge	32	244	2 x 320

Use Cases

We recommend memory-optimized instances for high performance databases, distributed memory caches, in-memory analytics, genome assembly and analysis, larger deployments of SAP, Microsoft SharePoint, and other enterprise applications.

EC2 Pricing Model

Free Usage Tier

On-Demand Instances

- Start and stop instances whenever you like, costs are rounded up to the nearest hour. (Worst price)

Reserved Instances

- Pay up front for one/three years in advance. (Best price)
- Unused instances can be sold on a secondary market.

Spot Instances

- Specify the price you are willing to pay, and instances get started and stopped without any warning as the market changes. (Kind of like Condor!)

<http://aws.amazon.com/ec2/pricing/>

Free Usage Tier

750 hours of EC2 running Linux, RHEL, or SLES t2.micro instance usage

750 hours of EC2 running Microsoft Windows Server t2.micro instance usage

750 hours of Elastic Load Balancing plus 15 GB data processing

30 GB of Amazon Elastic Block Storage in any combination of General Purpose (SSD) or Magnetic, plus 2 million I/Os (with Magnetic) and 1 GB of snapshot storage

15 GB of bandwidth out aggregated across all AWS services

1 GB of Regional Data Transfer

On-Demand Instance Prices

Linux

RHEL

SLES

Windows

Windows with SQL Standard

Windows with SQL Web

Region: US East (N. Virginia)

	vCPU	ECU	Memory (GiB)	Instance Storage (GB)	Linux/UNIX Usage
General Purpose - Current Generation					
t2.micro	1	Variable	1	EBS Only	\$0.013 per Hour
t2.small	1	Variable	2	EBS Only	\$0.026 per Hour
t2.medium	2	Variable	4	EBS Only	\$0.052 per Hour
m3.medium	1	3	3.75	1 x 4 SSD	\$0.070 per Hour
m3.large	2	6.5	7.5	1 x 32 SSD	\$0.140 per Hour
m3.xlarge	4	13	15	2 x 40 SSD	\$0.280 per Hour
m3.2xlarge	8	26	30	2 x 80 SSD	\$0.560 per Hour

Medium Utilization Reserved Instances

Linux	RHEL	SLES	Windows	Windows with SQL Standard	Windows with SQL Web
Region: US East (N. Virginia)					
1-Year Term			3-Year Term		
Upfront		Hourly	Upfront		Hourly
General Purpose - Current Generation					
t2.micro	\$42	\$0.006 per Hour	\$84	\$0.004 per Hour	
t2.small	\$84	\$0.012 per Hour	\$168	\$0.008 per Hour	
t2.medium	\$168	\$0.024 per Hour	\$336	\$0.016 per Hour	
m3.medium	\$181	\$0.027 per Hour	\$286	\$0.022 per Hour	
m3.large	\$362	\$0.055 per Hour	\$571	\$0.043 per Hour	
m3.xlarge	\$724	\$0.109 per Hour	\$1142	\$0.086 per Hour	
m3.2xlarge	\$1448	\$0.219 per Hour	\$2283	\$0.173 per Hour	

Spot Instance Pricing History



Product : **Linux/UNIX** ▾ Instance type: **c3.4xlarge** ▾ Date range : **1 month** ▾ Availability zone: **All zones** ▾



Availability zone	Price
 us-west-2a	\$0.1389
 us-west-2b	\$0.1390
 us-west-2c	\$0.1322

Date September 28, 2014 9:08:53 AM UTC-4

Surprisingly, you can't scale up that large.

Q: How many instances can I run in Amazon EC2?

You are limited to running up to 20 On-Demand Instances, purchasing 20 Reserved Instances, and requesting 5 Spot Instances per region. New AWS accounts may start with limits that are lower than the limits described here. Certain instance types are further limited per region as follows:

Instance Type	On-Demand Limit	Reserved Limit	Spot Limit
cg1.4xlarge	2	20	5
hi1.4xlarge	2	20	5
hs1.8xlarge	2	20	Not offered
cr1.8xlarge	2	20	5
g2.2xlarge	5	20	5

Simple Storage Service (S3)

A **bucket** is a container for objects and describes location, logging, accounting, and access control. A bucket can hold any number of **objects**, which are files of up to 5TB. A bucket has a name that must be **globally unique**.

Fundamental operations corresponding to HTTP actions:

- `http://bucket.s3.amazonaws.com/object`
- POST a new object or update an existing object.
- GET an existing object from a bucket.
- DELETE an object from the bucket
- LIST keys present in a bucket, with a filter.

A bucket has a **flat directory structure** (despite the appearance given by the interactive web interface.)

Easily Integrated into Web Applications

```
<form action="http://examplebucket.s3.amazonaws.com/" method="post" enctype="multipart/form-data">

<input type="input" name="key" value="user/user1/" />

<input type="hidden" name="acl" value="public-read" />
<input type="hidden" name="success_action_redirect"
      value="http://examplebucket.s3.amazonaws.com/successful_upload.html" />
...
<input type="text" name="X-Amz-Credential"
      value="AKIAIOSFODNN7EXAMPLE/20130806/us-east-1/s3/aws4_request" />
...
<input type="submit" name="submit" value="Upload to Amazon S3" /> </form>
```

<http://docs.aws.amazon.com/AmazonS3/latest/API/sigv4-post-example.html>

Bucket Properties

Versioning – If enabled, POST/DELETE result in the creation of new versions without destroying the old.

Lifecycle – Delete or archive objects in a bucket a certain time after creation or last access or number of versions.

Access Policy – Control **when and where** objects can be accessed.

Access Control – Control who **may** access objects in this bucket.

Logging – Keep track of how objects are accessed.

Notification – Be notified when failures occur.

Storage Pricing

Region:

US Standard



	Standard Storage	Reduced Redundancy Storage	Glacier Storage
First 1 TB / month	\$0.0300 per GB	\$0.0240 per GB	\$0.0100 per GB
Next 49 TB / month	\$0.0295 per GB	\$0.0236 per GB	\$0.0100 per GB
Next 450 TB / month	\$0.0290 per GB	\$0.0232 per GB	\$0.0100 per GB
Next 500 TB / month	\$0.0285 per GB	\$0.0228 per GB	\$0.0100 per GB
Next 4000 TB / month	\$0.0280 per GB	\$0.0224 per GB	\$0.0100 per GB
Over 5000 TB / month	\$0.0275 per GB	\$0.0220 per GB	\$0.0100 per GB

Request Pricing

Region:

US Standard



Pricing

PUT, COPY, POST, or LIST Requests

\$0.005 per 1,000 requests

Glacier Archive and Restore Requests

\$0.05 per 1,000 requests

Delete Requests

Free †

GET and all other Requests

\$0.004 per 10,000 requests

Glacier Data Restores

Free ‡

† No charge for delete requests of Standard or RRS objects. For objects that are archived to Glacier, there is a pro-rated charge of \$0.03 per gigabyte for objects deleted prior to 90 days. [Learn more.](#)

‡ Glacier is designed with the expectation that restores are infrequent and unusual, and data will be stored for extended periods of time. You can restore up to 5% of your average monthly Glacier storage (pro-rated daily) for free each month. If you choose to restore more than this amount of data in a month, you are charged a restore fee starting at \$0.01 per gigabyte. [Learn more.](#)

Data Transfer Pricing

The pricing below is based on data transferred "in" to and "out" of Amazon S3.

Region:	US Standard
Pricing	
Data Transfer IN To Amazon S3	
All data transfer in	\$0.000 per GB
Data Transfer OUT From Amazon S3 To	
Amazon EC2 in the Northern Virginia Region	\$0.000 per GB
Another AWS Region or Amazon CloudFront	\$0.020 per GB
Data Transfer OUT From Amazon S3 To Internet	
First 1 GB / month	\$0.000 per GB
Up to 10 TB / month	\$0.120 per GB
Next 40 TB / month	\$0.090 per GB
Next 100 TB / month	\$0.070 per GB

Elastic Block Store

An EBS volume is a **virtual disk** of a fixed size with a block read/write interface. It can be **mounted** as a filesystem on a running EC2 instance where it can be **updated incrementally**. Unlike an instance store, an EBS volume is **persistent**.

(Compare to an S3 object, which is essentially a file that must be accessed in its entirety.)

Fundamental operations:

- CREATE a new volume (1GB-1TB)
- COPY a volume from an existing EBS volume or S3 object.
- MOUNT on one instance at a time.
- SNAPSHOT current state to an S3 object.

Amazon EBS Volume Types

Volume Type	EBS General Purpose (SSD)	EBS Provisioned IOPS (SSD)	EBS Magnetic
Use Cases	Boot volumes Small to Med DBs Dev and Test	I/O intensive Relational DBs NoSQL DBs	Infrequent Data Access
Storage Media	SSD-backed	SSD-backed	Magnetic disk-backed
Max Volume Size	1TB	1TB	1TB
Max IOPS/volume	3,000 (burst)	4,000	40 - 200
Max throughput/volume	128MBps	128MBps	40 - 90MBps
Max IOPS/instance	48,000	48,000	48,000
Max throughput/instance	800MBps	800MBps	800MBps
API Name	gp2	io1	standard
Price*	\$.10/GB - Month	\$.125/GB - Month \$.065/provisioned IOPS	\$.05/GB - Month \$.05/million I/O

Amazon EBS Pricing

Cr

With Amazon EBS, you only pay for what you use. The pricing for Amazon EBS volumes is listed below.

Region:

US East (N. Virginia)



Amazon EBS General Purpose (SSD) volumes

- \$0.10 per GB-month of provisioned storage

Amazon EBS Provisioned IOPS (SSD) volumes

- \$0.125 per GB-month of provisioned storage
- \$0.065 per provisioned IOPS-month

Amazon EBS Magnetic volumes

- \$0.05 per GB-month of provisioned storage
- \$0.05 per 1 million I/O requests

Amazon EBS Snapshots to Amazon S3

- \$0.095 per GB-month of data stored

Use Glacier for Cold Data

Glacier is structured like S3: a **vault** is a container for an arbitrary number of **archives**.

Policies, accounting, and access control are associated with vaults, while an archive is a single object.

However:

- All operations are asynchronous and notified via SNS.
- Vault listings are updated once per day.
- Archive downloads may take up to four hours.
- Only 5% of total data can be accessed in a given month.

Pricing:

- Storage: \$0.01 per GB-month
- Operations: \$0.05 per 1000 requests
- Data Transfer: Like S3, free within AWS.

S3 Policies can be set up to automatically move data into Glacier.

Durability

Amazon claims about S3:

- Amazon S3 is designed to sustain the concurrent loss of data in two facilities, e.g. 3+ copies across multiple available domains.
- 99.999999999% durability of objects over a given year.

Amazon claims about EBS:

- Amazon EBS volume data is replicated across multiple servers in an Availability Zone to prevent the loss of data from the failure of any single component.
- Volumes <20GB modified data since last snapshot have an annual failure rate of 0.1% - 0.5%, resulting in complete loss of the volume.

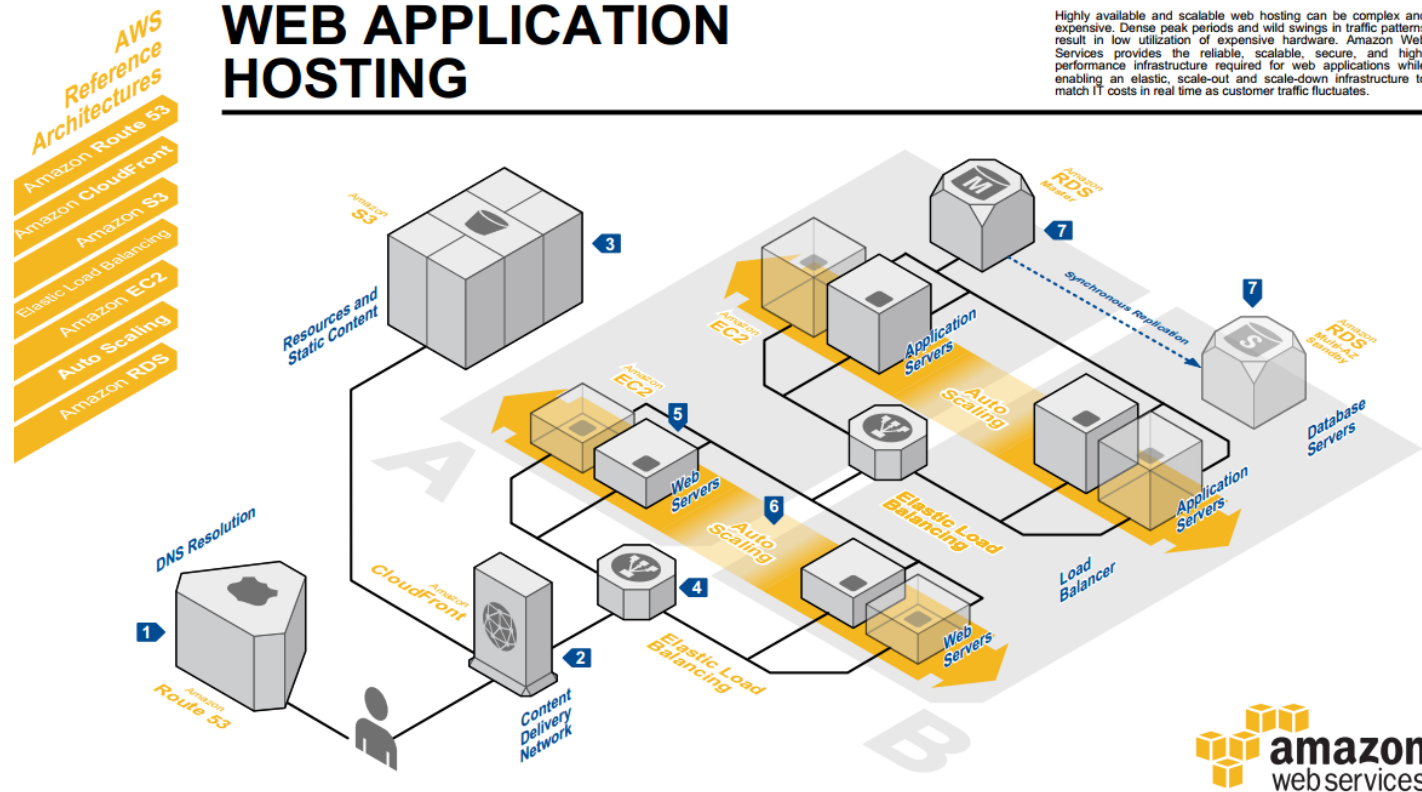
Amazon claims about Glacier is the same as S3:

- Amazon S3 is designed to sustain the concurrent loss of data in two facilities, e.g. 3+ copies across multiple available domains PLUS periodic internal integrity checks.
- 99.999999999% durability of objects over a given year.

Architecture Center

Ideas for constructing large scale infrastructures using AWS:

<http://aws.amazon.com/architecture/>



Command Line Setup

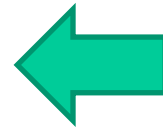
Go to your profile menu (your name) in the upper right hand corner, select “Security Credentials” and “Continue to Security Credentials”

Select “Access Keys”

Select “New Access Key” and save the generated keys somewhere.

Edit ~/.aws/config and set it up like this:

```
[default]
output = json
region = us-west-2
aws_access_key = XXXXXX
aws_secret_access_key = YYYYYYYYYYYY
```



Note the syntax here is different from how it was given in the web console!

AWSAccessKey=XXXXXX

AWSSecretAccessKey=YYYYYYYYYY

No

ances

S3 Command Line Examples

```
aws s3 mb s3://bucket
...    cp localfile s3://bucket/key
        mv s3://bucket/key s3://bucket/newname
        ls s3://bucket
        rm s3://bucket/key
        rb s3://bucket
```

```
aws s3 help
aws s3 ls help
```

EC2 Command Line Examples

```
aws    ec2    describe-instances
        run-instances --image-id ami-xxxxx -- count 1
                        --instance-type t1.micro --key-name keyfile
        stop-instances --instance-id i-xxxxxx
```

```
aws    ec2    help
```

```
aws    ec2    start-instances help
```

Warmup: Get Started with Amazon

Skim through the AWS documentation.

Sign up for AWS at <http://aws.amazon.com>

(Skip the IAM management for now)

Apply the service credit you received by email.

Create and download a Key-Pair, save it in your home directory.

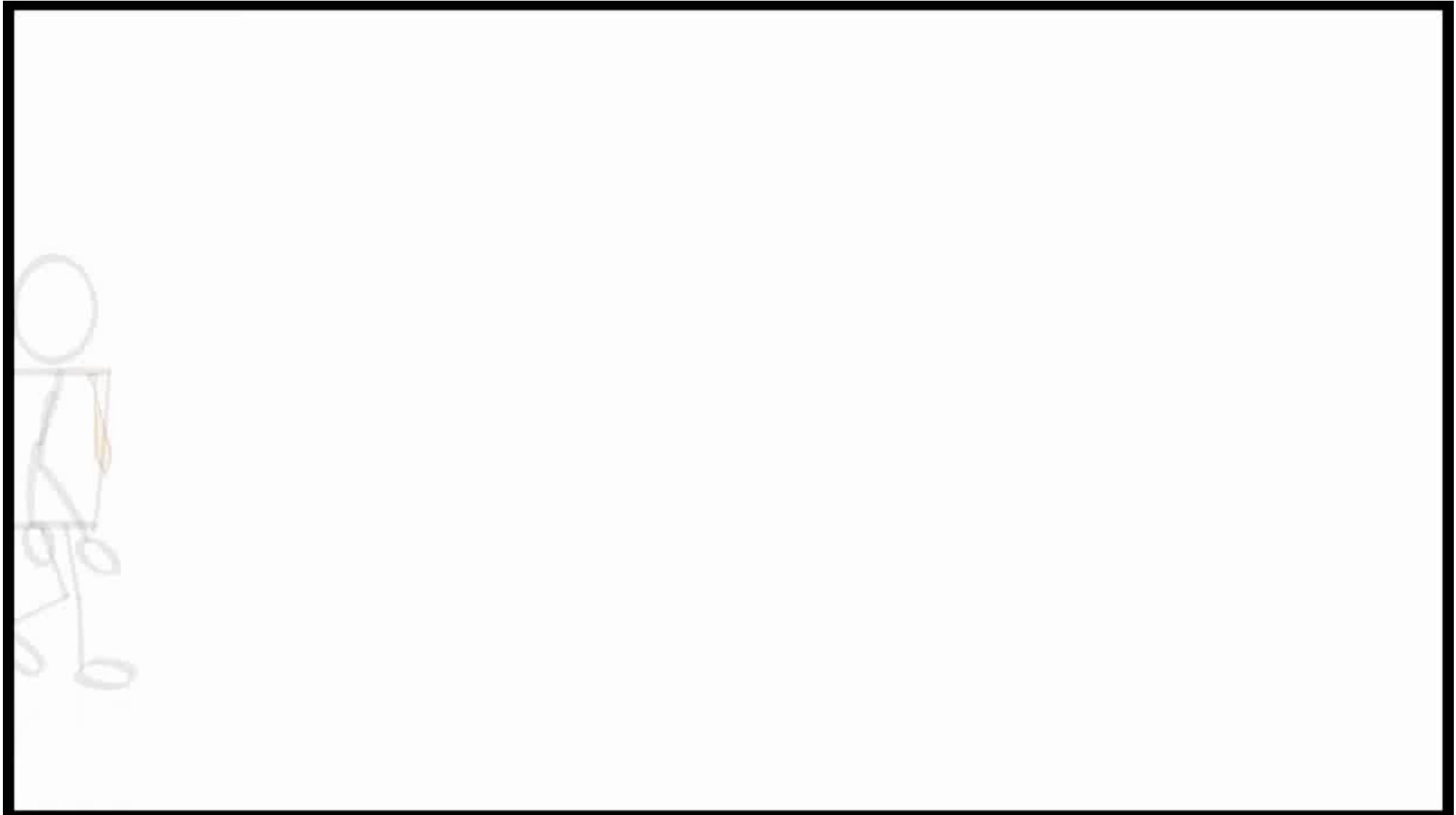
Create a VM via the AWS Console

Connect to your newly-created VM like this:

- `ssh -i my-aws-keypair.pem ec2-user@ip-address-of-vm`

Create a bucket in S3 and upload/download some files.

Demo Time



SLAs, QoS and monitoring

AWS Service Level Agreements (SLAs)

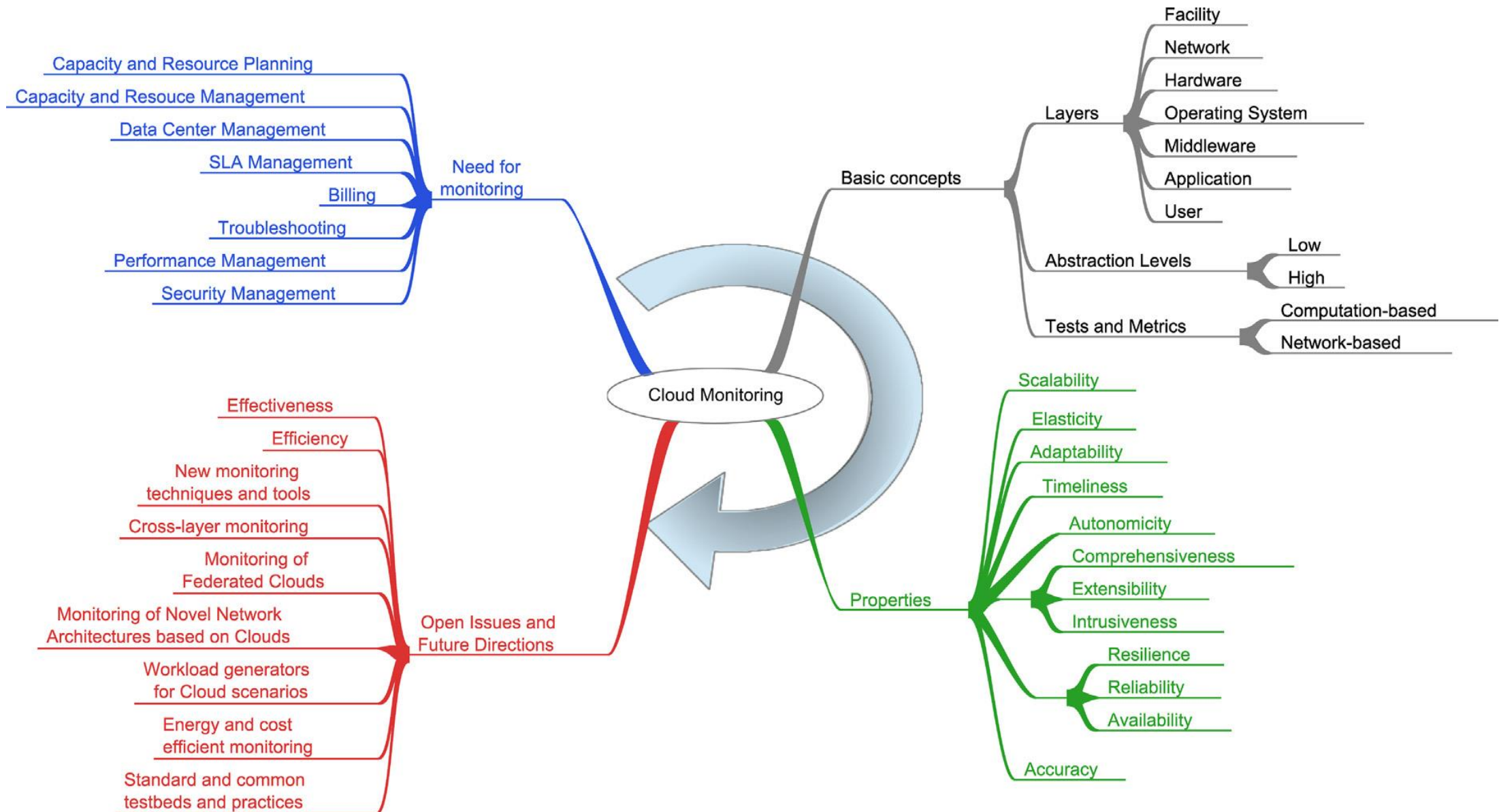
AWS commits to offer Service Level Agreements (SLAs) and publish Service Level Objectives (SLOs) for all paid, generally available services

<https://aws.amazon.com/legal/service-level-agreements>

Quality of Service (QoS)

Quality of service (QoS) is the description or measurement of the overall performance of a service, such as a telephony or computer network or a cloud computing service, particularly the performance seen by the users of the network.

Monitoring Generations



Monitoring Platform Comparisons

Platform	Scalability	Elasticity	Adaptability	Timeliness	Autonomicity	Comprehensiveness	Extensibility	Intrusiveness	Resilience	Reliability	Availability	Accuracy
CloudWatch [95]		✓		✓			✓					
AzureWatch [137]	✓		✓		✓		✓					
CloudKick [96]	✓		✓									
CloudStatus [48]				✓								
Nimsoft [97]	✓					✓						
Monitis [99]						✓						
LogicMonitor [100]	✓	✓				✓						
Aneka [101]	✓	✓										
GroundWork [129]						✓						

Open Source Platform Monitoring Comparisons

Platform	Scalability	Elasticity	Adaptability	Timeliness	Autonomicity	Comprehensiveness	Extensibility	Intrusiveness	Resilience	Reliability	Availability	Accuracy
Nagios [104]							✓					
OpenNebula [105]	✓		✓									
CloudStack ZenPack [109]				✓								
Nimbus [110]					✓							
PCMONS [111]							✓					
DARGOS [128]			✓				✓	✓				
Hyperic-HQ [138]	✓					✓						
Sensu [139]		✓					✓					

Research Directories on Cloud

